Mechanical Energy Turning Into Heat

Student Notes + Worked Problems + Practice

Main formulas

$$W = JQ, \qquad J \approx 4.2 \text{ J/cal}$$

$$\boxed{f \times (\text{mechanical energy lost}) = mc\Delta T \; [+mL]}$$

Use $mc\Delta T$ when temperature changes. Add mL only if a phase change occurs.

1. Mechanical Energy Can Become Heat

In these problems, there may be no hot body at all. A bullet stops, water falls, two bodies collide, or a block slides with friction. The lost mechanical energy can appear as heat.

$$\boxed{\text{Mechanical energy lost} = \text{thermal energy gained}}$$

Typical mechanical-energy terms are:

$$\frac{1}{2}mv^2, \qquad mgh, \qquad KE_i - KE_f.$$

1.1 Mechanical equivalent of heat

Joule showed that work and heat are two forms of the same energy. The conversion is

$$W = JQ, \qquad J \approx 4.2 \text{ J/cal}.$$

Therefore,

$$1 \text{ cal} \approx 4.2 \text{ J}.$$

Worked Example 1 - Joule to calorie conversion

420 J of mechanical energy is completely converted into heat. Find the heat in calories.

$$Q = \frac{W}{J} = \frac{420}{4.2} = 100 \text{ cal}$$

$$\boxed{Q = 100 \text{ cal}}$$

2. The Conversion Template

If only a fraction f of the lost mechanical energy heats the body,

$$\boxed{fE_{\text{lost}} = mc\Delta T + mL}$$

The mL term is present only if the body melts or undergoes another phase change.

Reading the fraction f correctly
If "50% of the kinetic energy heats the bullet," then $f = 0.50$.
If "25% of the energy is absorbed by the obstacle," then 75% heats the bullet, so $f = 0.75$.

2.1 Bullet that just melts

For a bullet of speed v that stops and just melts,

$$f\left(\frac{1}{2}mv^2\right) = mc(T_m - T_i) + mL_f.$$

The mass cancels:

$$\boxed{f\left(\frac{1}{2}v^2\right) = c(T_m - T_i) + L_f}.$$

Worked Example 2 - Bullet melts

A lead bullet starts at 27°C and just melts at 327°C. Half of its kinetic energy heats the bullet. Take

$$c = 125 \text{ J kg}^{-1}\text{K}^{-1}, \qquad L_f = 2.5 \times 10^4 \text{ J kg}^{-1}.$$

Temperature rise:

$$\Delta T = 327 - 27 = 300^\circ\text{C}.$$

Heat needed per kilogram:

$$c\Delta T + L_f = (125)(300) + 2.5 \times 10^4 = 62500 \text{ J/kg}.$$

Now

$$0.5\left(\frac{1}{2}v^2\right) = 62500$$

$$0.25v^2 = 62500 \quad \Rightarrow \quad v^2 = 250000$$

$$\boxed{v = 500 \text{ m/s}}$$

3. Falling Body and Waterfall Problems

If a body falls through height h and all the lost potential energy becomes heat,

$$mgh = mc\Delta T.$$

The mass cancels:

$$\boxed{\Delta T = \frac{gh}{c}}.$$

Worked Example 3 - Waterfall

Water falls through a height of 200 m. Assume all the lost potential energy heats the water. Take

$$g = 10 \text{ m/s}^2, \qquad c = 4200 \text{ J kg}^{-1}\text{K}^{-1}.$$

$$\Delta T = \frac{gh}{c} = \frac{(10)(200)}{4200}$$

$$\boxed{\Delta T \approx 0.48 \text{ K}}$$

The result does not depend on the mass of water.

4. Inelastic Collision: Lost Kinetic Energy Becomes Heat

For a perfectly inelastic collision, first use momentum conservation to find the common final velocity. Then find the kinetic energy that disappears:

$$Q = KE_i - KE_f.$$

If all of this becomes heat in the bodies,

$$\boxed{KE_i - KE_f = m_{\text{total}} c \Delta T}.$$

Worked Example 4 - Collision

Two spheres of masses 15 kg and 25 kg move towards each other at 10 m/s and 30 m/s and stick together. Take

$$c = 31 \text{ cal kg}^{-1}{}^{\circ}\text{C}^{-1}.$$

Choose the 25 kg sphere's direction as positive.

$$p = (25)(30) - (15)(10) = 600 \text{ kg m/s}$$

$$v = \frac{600}{15 + 25} = 15 \text{ m/s}$$

Initial kinetic energy:

$$KE_i = \frac{1}{2}(15)(10^2) + \frac{1}{2}(25)(30^2) = 12000 \text{ J}$$

Final kinetic energy:

$$KE_f = \frac{1}{2}(40)(15^2) = 4500 \text{ J}$$

Hence,

$$Q = 12000 - 4500 = 7500 \text{ J}.$$

Convert specific heat:

$$c = (31)(4.2) = 130.2 \text{ J kg}^{-1}{}^{\circ}\text{C}^{-1}.$$

Then

$$7500 = (40)(130.2)\Delta T$$

$$\boxed{\Delta T \approx 1.44^{\circ}\text{C}}$$

Collision traps
Opposite directions require opposite momentum signs. After the bodies stick, the heat warms the **total mass**, not just one body.

5. Sliding Down a Rough Incline at Constant Speed

If a block slides down a rough incline at constant speed, then

$$\Delta KE = 0.$$

So the loss in gravitational potential energy becomes heat. If the block moves a distance L along an incline of angle θ,

$$h = L\sin\theta.$$

Therefore,

$$Q = mgL\sin\theta.$$

If all this heat stays in the block,

$$mgL\sin\theta = mc\Delta T.$$

Hence,

$$\boxed{\Delta T = \frac{gL\sin\theta}{c}}.$$

Worked Example 5 - Copper block on a rough incline

A 200 g copper block slides 60 cm down a rough 37° incline at constant speed. All the lost mechanical energy heats the block. Take

$$c = 420\ \text{J kg}^{-1}\text{K}^{-1}, \qquad g = 10\ \text{m/s}^2, \qquad \sin 37^\circ \approx 0.6.$$

Using

$$\Delta T = \frac{gL\sin\theta}{c},$$

$$\Delta T = \frac{(10)(0.60)(0.6)}{420}$$

$$\boxed{\Delta T \approx 8.6 \times 10^{-3}\ ^\circ\text{C}}$$

The phrase **constant speed** tells us that kinetic energy does not change.

6. Quick Revision

Situation	Equation / idea
Joule-calorie conversion	$W = JQ$, $J \approx 4.2$ J/cal
Bullet stops and heats	$f(\frac{1}{2}mv^2) = mc\Delta T$
Bullet just melts	$f(\frac{1}{2}mv^2) = mc\Delta T + mL_f$
Falling body / waterfall	$mgh = mc\Delta T$, so $\Delta T = gh/c$
Inelastic collision	$KE_i - KE_f = m_{\text{total}}c\Delta T$
Constant-speed sliding	loss in PE = heat; $\Delta KE = 0$

One habit for every question
First identify **which mechanical energy is lost**. Then identify **what fraction reaches the body**. Finally equate that energy to the heat gained.

7. Practice Problems

Problem 1 - Conversion
840 J of work is completely converted into heat. Find the heat produced in calories. Take $J = 4.2$ J/cal.

Problem 2 - Bullet heating and melting
A lead bullet starts at 27°C and just melts at 327°C. Half of its kinetic energy is retained by the bullet. Take $c = 125$ J/kg-K and $L_f = 2.5 \times 10^4$ J/kg. Find its speed.

Problem 3 - Waterfall
Water falls through 420 m. Assume all the lost potential energy heats the water. Take $g = 10$ m/s^2 and $c = 4200$ J/kg-K. Find the rise in temperature.

Problem 4 - Ball drop and rebound
A ball is dropped from 2.0 m and rebounds to 1.5 m. If 40% of the mechanical energy lost becomes thermal energy in the ball, find the temperature rise. Take $c = 800$ J/kg-K and $g = 10$ m/s^2.

Problem 5 - Rough incline
A copper block slides down a rough 37° incline at constant speed through 0.60 m. All the lost potential energy heats the block. Take $c = 420$ J/kg-K, $g = 10$ m/s^2, $\sin 37° = 0.6$. Find the temperature rise.

8. Answers to Practice Problems

1. 200 cal
2. 500 m/s
3. 1.0 K
4. Mechanical energy lost per kg $= g(2.0 - 1.5) = 5$ J/kg. Heat retained per kg $= 0.4(5) = 2$ J/kg. Hence $\Delta T = 2/800 = 2.5 \times 10^{-3}$ K.
5. 8.6×10^{-3} °C